Nikhil Kumar

Shimla, Himachal Pradesh, India
Email: nikhilsoni8557@gmail.com
LinkedIn: linkedin.com/in/nikhil-kumar-3197641b7
GitHub: github.com/Nikhil8557

FOCUS

I am currently developing a within-host genomic foundation model that anticipates viral mutation patterns from host samples and maps the trajectory of viral spread, accounting for drug introductions and host immune responses. I am also working to integrate this framework with Next-Generation Sequencing (NGS) wet-lab pipelines to create an end-to-end immune monitoring system. Additionally, I focus on building AI-driven guardrails to prevent the misuse or dual-use exploitation of this research. I am dedicated to AI safety because if I am developing AI models, it is my responsibility to keep them safe. I am seeking high-impact research or fellowship roles, with a preference for remote opportunities. I thrive in convergence science, an interdisciplinary approach that combines diverse methodologies across fields to solve core challenges.

CORE COMPETENCIES & TECHNICAL EXPERTISE

Genomic AI & Bio-ML:	Within-Host Mutation Trajectory Forecasting, Topologically Associating Domain (TAD) Mapping (Hi-C), Spatio-Temporal 3D Chromatin Modeling, miRNA Target Site Identification, Multimodal NGS Data Integration.
AI Safety & Alignment:	Non-Agentic AI Safety Guardrails, Dual-Use Research Governance, Mathematical Safety Guarantees, Biological Biorisk Mitigation, Scientist AI Research Frameworks.
Machine Learning & Systems:	Convolutional Neural Networks (CNNs), Transformers (Unified Correspondence Architecture), Self-Supervised Learning (SSL), Python, PyTorch, Linux / Shell Scripting, Multimodal Embedding Spaces.
Theoretical Research:	Particle Physics (PARMANU / CERN), Quantum Statistical Mechanics, Integer Partitions & Higher-Dimensional Geometry, Topology, Number Theory.

RESEARCH EXPERIENCE & TECHNICAL VENTURES

VIRAM *(Viral Retention and Mutation Analysis AI)* — **2022 - Present**
Lead Researcher & Founder — Independent Research

- Architected and developed a within-host genomic foundation model that anticipates viral mutation trajectories from host samples and maps viral spread under drug and immune pressure.
- Engineered multimodal data pipelines combining Next-Generation Sequencing (NGS) data with unified embedding spaces and self-supervised learning (SSL) to resolve batch effect disparities.
- Secured competitive independent research grant funding from **Emergent Ventures** to scale the predictive framework into an end-to-end immune monitoring system.
- Designed robust AI-driven guardrails to prevent the dual-use exploitation of genomic mutation forecasting models.

CSIR-IHBT *(CSIR - Institute of Himalayan Bioresource Technology)* — **June 2026 - July 2026**
Research Trainee / Intern - Bioinformatics Laboratory — Palampur, India

- **Spatio-Temporal Chromatin Modeling:** Tracked 3D chromatin architecture evolution over time using Unified Correspondence Transformers across 2D genomic maps and 3D structures to achieve 2D-to-3D structural alignment.
- **TAD Mapping:** Conducted Topologically Associating Domain (TAD) discovery through Hi-C heat mapping, evaluating novel state-of-the-art architectures to improve genomic data quality.
- **Site Accessibility Modeling:** Developed a CNN-based deep learning model for miRNA target site identification on imbalanced biological datasets using global max pooling; co-authored paper presented at ICCBB (Japan).

SAGARai — **2025 - Present**
Technical Founder — Venture

- Founded an AI venture focused on developing advanced multimodal computational frameworks for localized tracking and intensity forecasting.
- Admitted into the **Microsoft Founders Hub**, obtaining enterprise cloud infrastructure, GPU compute resources, and technical support.

HONORS, FELLOWSHIPS & COMMENDATIONS

NASA VDP Commendation	Recognized by the **NASA Vulnerability Disclosure Program** for identifying critical vulnerabilities and contributing to cybersecurity engineering.	**2026**
Emergent Ventures Grantee	Awarded competitive independent research grant funding for breakthrough architecture in viral mutation forecasting and genomic AI (VIRAM).	**2024**
PMO Official Commendation	Received official Letter of Appreciation / Commendation from the **Prime Minister's Office (PMO), India**, for technical excellence and impact.	**2024**
CERN BL4S Fellow	Selected for the international Beamline for Schools physics fellowship program at **CERN**; proposed particle physics research paper under PARMANU (Geneva).	**2023 - 2024**
Microsoft Founders Hub Member	Awarded technical equity, enterprise cloud credits, and infrastructure support to scale AI model deployment.	**2025**

AI SAFETY, ALIGNMENT & BIOSECURITY ENGAGEMENTS

- **Global Expert Dialogue on AI Safety:** Engaging in academic correspondence with leading global experts (including Turing Award laureate Yoshua Bengio) to explore non-agentic AI architectures, Scientist AI frameworks, and mathematical safety guarantees for deep neural networks.
- **BlueDot Impact Biosecurity Cohort (2026 - Present):** Selected for intensive biosecurity training focusing on global catastrophic biological risks, dual-use research governance, biorisk mitigation strategies, and AI guardrails.
- **Cluster AI Safety Initiatives:** Organized and led cluster-level AI Safety and AI4Good programs in collaboration with Microsoft to promote responsible AI development across ambassador networks.

LEADERSHIP & COMMUNITY OUTREACH

Microsoft Learn Student Ambassador - *Beta Tier Lead* **2022 - 2025**

- Founded and led the region's first Microsoft Learn Student Ambassador and Hack Club chapter, growing membership to 150+ students across campus.
- Organized and facilitated 7+ technical workshops and developer bootcamps focusing on cloud architectures, AI safety, and software engineering.

GeeksforGeeks - *Campus Ambassador* **2022 - 2024**

- Spearheaded technical education initiatives on campus, mentoring peers in data structures, algorithms, system concepts, and competitive programming.

Your Big Year & Community Activism - *International Fellow & Advocate* **2023 - Present**

- Selected as an international fellow collaborating on global youth leadership, cross-cultural impact frameworks, and community literacy initiatives.
- Organized local climate literacy campaigns and animal welfare community drives.

EDUCATION & SPECIALIZED PROGRAMS

Kendriya Vidyalaya, Palampur (CBSE) **Palampur, Himachal Pradesh**

Senior Secondary & Secondary Education

- **Class 12 - 2025:** Focus on Physics, Chemistry, and Mathematics | **Aggregate Score: 87%**
- **Class 10 - 2023:** Secondary School Examination | **Aggregate Score: 91%**
  President: PM SHRI Innovation Council & Science Club

Selected Industry & Academic Programs: **2020 - 2022**

Deloitte Quantum Technology Program • HSBC Global Banking & Markets Experience • Goldman Sachs Engineering Program • Microsoft Engineering Asia Experience • Harvard Medical School Pediatrics Program.